HARD Mock Set 3 — Accenture Technical (Above-Level)

Level: Harder than the 10 uploaded sets. Unique questions — nothing here repeats Sets 1, 2, 4 or 5.

Format: 50 questions · Time: 50 minutes, closed book · No negative marking.

Section	Area	Qs
A	MS Office & Common Applications	8
B	Networking, Security & Cloud	12
C	CS Core — DSA / DBMS / OOP / HTML-CSS-JS	12
D	Pseudocode & Code Tracing	14
E	Applied — SQL & Code Output	4

Section A — MS Office & Common Applications (8)

Q1. A1:A5 = 5, 20, 10, 15, 8 and B1:B5 = 1, 2, 3, 4, 5. What does `=SUMPRODUCT((A1:A5>10) * B1:B5)` return?

A) 15

B) 8

C) 6

D) 3

Q2. What does `=TEXT(0.75, "0.0%")` display in Excel?

A) `0.8%`

B) `75.0%`

C) `75%`

D) `.75`

Q3. What does `=LEN(TRIM(" Hello World "))` return?

A) 15

B) 13

C) 10

D) 11

Q4. `=IF(A1 > 50, B1 * 0.1, B1 * 0.05)` with A1 = 50 and B1 = 200. Result?

A) 10

B) 20

C) 5

D) 0

Q5. In MS Word, what is a **section break** used for?

A) Splitting one document into two files

B) Inserting a page number

C) Applying a different header, page numbering or page orientation **within** the same document

D) Ending a paragraph

Q6. Which PowerPoint view shows all slides as thumbnails so you can reorder and set timings?

A) Reading View

B) Slide Sorter View

C) Notes Page View

D) Outline View

Q7. You must email 50 recipients without letting them see each other's addresses. Which field?

A) To

B) Cc

C) Reply-To

D) Bcc

Q8. Which Excel chart type best shows the **relationship/correlation between two numeric variables**?

A) Scatter (XY) chart

B) Pie chart

C) Column chart

D) Treemap

Section B — Networking, Security & Cloud (12)

Q9. A subnet must support **100 hosts**. Smallest prefix?

A) /24

B) /25

C) /26

D) /27

Q10. `10.0.0.0/24` is divided into 4 equal subnets. Which is the **second** subnet?

A) 10.0.0.0/26

B) 10.0.0.128/26

C) 10.0.0.64/26

D) 10.0.0.32/26

Q11. Which is a **valid host address** in `192.168.1.0/28`?

A) 192.168.1.0

B) 192.168.1.15

C) 192.168.1.16

D) 192.168.1.14

Q12. The IPv6 **loopback** address is:

A) `::1`

B) `FE80::1`

C) `FF02::1`

D) `2001:db8::1`

Q13. Which prefix is reserved for IPv6 **link-local** addresses?

A) `2000::/3`

B) `FC00::/7`

C) `FE80::/10`

D) `FF00::/8`

Q14. Which protocol replaces ARP for address resolution in IPv6?

A) RARP

B) DHCPv6

C) IGMP

D) NDP (Neighbor Discovery, using ICMPv6)

Q15. Which file-transfer protocol runs **over SSH**?

A) FTPS

B) SFTP

C) TFTP

D) SMB

Q16. Which statement about TCP/UDP is **FALSE**?

A) UDP is connection-oriented

B) TCP delivers bytes in order

C) TCP uses a three-way handshake

D) UDP has no built-in congestion control

Q17. Which device can filter traffic using **IP addresses and port numbers** (Layers 3–4)?

A) Hub

B) Layer-2 switch

C) Repeater

D) Firewall

Q18. Automatically adding instances at peak load and removing them afterwards is called:

A) Provisioning

B) Orchestration

C) Auto-scaling / elasticity

D) Load-balancing

Q19. The most effective control against **credential stuffing** (replayed leaked passwords) is:

A) Password complexity rules

B) Multi-factor authentication (MFA)

C) Longer session timeouts

D) Antivirus

Q20. A **Type 1 (bare-metal) hypervisor**:

A) Runs directly on the host hardware, with no host OS beneath it

B) Runs as an application on top of a host OS

C) Is another name for a container runtime

D) Cannot run more than one VM

Section C — CS Core: DSA / DBMS / OOP / HTML-CSS-JS (12)

Q21. Which operation is **O(1) amortised** but O(n) in the worst case?

A) Search in a balanced BST

B) Deletion from a hash table

C) Push onto a linked-list stack

D) Appending to a dynamically-resizing array

Q22. Which sort is optimal for an **almost-sorted** array?

A) Heap sort

B) Insertion sort

C) Selection sort

D) Quick sort with last-element pivot

Q23. Which structure checks whether parentheses in an expression are **balanced**?

A) Stack

B) Queue

C) Binary tree

D) Graph

Q24. A **complete** binary tree has 31 nodes. How many leaves does it have?

A) 15

B) 31

C) 16

D) 8

Q25. Which statement about 3NF and BCNF is TRUE?

A) Every 3NF relation is in BCNF

B) BCNF and 3NF are identical

C) BCNF allows transitive dependencies

D) Every BCNF relation is in 3NF, but a 3NF relation need not be in BCNF

Q26. `R(A,B,C,D)` with FDs `A`→`B`, `B`→`C`, `C`→`D`. What is the candidate key?

A) `{A,B}`

B) `{A}`

C) `{A,B,C}`

D) `{D}`

Q27. Table `Orders(id, cust, amount)` → (1,A,100), (2,B,250), (3,A,300), (4,C,50), (5,B,200). Which customer appears **first**?

```
SELECT cust, COUNT(*) c, SUM(amount) s
FROM Orders
GROUP BY cust
HAVING SUM(amount) >= 300
ORDER BY c DESC, s DESC;
```

A) C

B) A

C) No rows returned

D) B

Q28. Which ACID property guarantees concurrent transactions behave as if executed **serially**?

A) Atomicity

B) Isolation

C) Durability

D) Consistency

Q29. Java — what is printed by `new B();`?

```
class A { A() { show(); } void show() { System.out.print("A"); } }
class B extends A { void show() { System.out.print("B"); } }
```

A) A

B) AB

C) B

D) Compile-time error

Q30. Which HTML5 element is meant to contain the **primary navigation links** of a page?

A) `<header>`

B) `<menu>`

C) `<div id="nav">`

D) `<nav>`

Q31. JavaScript:

```
const arr = [1, 2, 3];
console.log(arr.map(x => x * 2).filter(x => x > 2));
```

A) `[2, 4, 6]`

B) `[4, 6]`

C) `[6]`

D) `[2, 4]`

Q32. Minimum number of bits required to represent **500 distinct values**?

A) 9

B) 8

C) 10

D) 16

Section D — Pseudocode & Code Tracing (14)

Q33.

```
a ← 0; b ← 0
FOR i ← 1 TO 5
    IF i MOD 2 = 0
        a ← a + i
    ELSE
        b ← b + i
    END IF
END FOR
PRINT a, b
```

A) 15 0

B) 5 10

C) 6 9

D) 9 6

Q34.

```
FUNCTION SD(n)
    IF n < 10
        RETURN n
    RETURN n MOD 10 + SD(n DIV 10)
END FUNCTION
PRINT SD(4567)
```

A) 4

B) 10

C) 4567

D) 22

Q35. (x is a **global** variable)

```
x ← 10
FUNCTION f(n)
    x ← x + n
    RETURN x
END FUNCTION
r1 ← f(3)
r2 ← f(5)
PRINT r1 + r2
```

A) 23

B) 31

C) 18

D) 26

Q36. How many times does a character repeat consecutively in `"mississippi"`?

```
s ← "mississippi"
count ← 0
FOR i ← 0 TO LENGTH(s) - 2
    IF s[i] = s[i + 1]
        count ← count + 1
    END IF
END FOR
PRINT count
```

A) 3

B) 4

C) 2

D) 5

Q37.

```
i ← 1; j ← 10; c ← 0
WHILE i < j
    i ← i + 2
    j ← j - 1
    c ← c + 1
END WHILE
PRINT c, i, j
```

A) 2 5 8

B) 4 9 6

C) 5 11 5

D) 3 7 7

Q38.

```
A ← [1, 2, 3, 4, 5, 6]
i ← 0; j ← 5
WHILE i < j
    temp ← A[i]
    A[i] ← A[j]
    A[j] ← temp
    i ← i + 1
    j ← j - 1
END WHILE
PRINT A[1], A[4]
```

A) 2 5

B) 5 2

C) 6 1

D) 1 6

Q39.

```
FUNCTION F(n)
    IF n = 0 RETURN 0
    IF n = 1 RETURN 1
    RETURN 2 * F(n-1) + F(n-2)
END FUNCTION
PRINT F(4)
```

A) 5

B) 24

C) 12

D) 8

Q40. Selection sort on `A ← [64, 25, 12, 22, 11]`, counting only executed swaps. How many swaps?

A) 3

B) 5

C) 4

D) 2

Q41.

```
n ← 100; c ← 0
WHILE n > 1
    n ← n DIV 3
    c ← c + 1
END WHILE
PRINT c, n
```

A) 5 0

B) 3 3

C) 4 3

D) 4 1

Q42.

```
FUNCTION firstUpper(s)
    FOR i ← 0 TO LENGTH(s) - 1
        IF s[i] is an uppercase letter
            RETURN i
        END IF
    END FOR
    RETURN -1
END FUNCTION
PRINT firstUpper("acCenture")
```

A) 0

B) 2

C) 3

D) -1

Q43. 2×2 matrix multiplication:

```
M ← [[1,2],[3,4]]
N ← [[5,6],[7,8]]
FOR i ← 0 TO 1
    FOR j ← 0 TO 1
        C[i][j] ← M[i][0] * N[0][j] + M[i][1] * N[1][j]
    END FOR
END FOR
PRINT C[0][0], C[1][1]
```

A) 19 23

B) 17 50

C) 19 50

D) 23 50

Q44.

```
i ← 0; s ← 0
WHILE i < 10
    i ← i + 1
    IF i MOD 4 = 0
        CONTINUE
    END IF
    s ← s + i
END WHILE
PRINT s
```

A) 43

B) 55

C) 47

D) 31

Q45.

```
s ← "abxcba"
n ← LENGTH(s)
ok ← TRUE
FOR i ← 0 TO (n DIV 2) - 1
    IF s[i] <> s[n - 1 - i]
        ok ← FALSE
        BREAK
    END IF
END FOR
PRINT ok
```

A) TRUE (loop never runs)

B) TRUE (all compared pairs match)

C) FALSE at i = 0

D) FALSE (mismatch detected at i = 2)

Q46. Tower of Hanoi move count:

```
FUNCTION moves(n)
    IF n = 1 RETURN 1
    RETURN 2 * moves(n-1) + 1
END FUNCTION
PRINT moves(5)
```

A) 15

B) 31

C) 16

D) 63

Section E — Applied: SQL & Code Output (4)

Salaries table for Q47: `Emp(id, salary)` → (1, 70000), (2, 70000), (3, 60000), (4, 50000), (5, 45000)

Q47. What does this return?

```
SELECT MAX(salary) FROM Emp
WHERE salary < (SELECT MAX(salary) FROM Emp);
```

A) 70000

B) NULL

C) 60000

D) 50000

Q48. Which statement about `DELETE`, `TRUNCATE` and `DROP` is TRUE?

A) `TRUNCATE TABLE` removes all rows, is minimally logged, and **cannot** take a `WHERE` clause

B) `DELETE` without `WHERE` is always faster than `TRUNCATE`

C) `DROP TABLE` deletes rows but keeps the table structure

D) `TRUNCATE` is DML and can be rolled back in every DBMS

Q49. Java:

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++) {
    arr[i] = arr[arr.length - 1 - i];
}
System.out.print(arr[0] + "" + arr[1] + "" + arr[2]);
```

A) `123`

B) `321`

C) `111`

D) `323`

Q50. HTML/JS — what is rendered on the page?

```
<p id="demo">Hello</p>
<script>
  document.getElementById('demo').innerHTML = '<b>Hi</b>';
</script>
```

A) The literal text `<b>Hi</b>`

B) The word **Hi** in bold

C) Nothing — you must use `textContent`

D) `Hello` unchanged, because the script runs before the element exists

ANSWER KEY

Do not look here until you have finished all 50 questions.

— Answer Key for Set 3

Quick grid

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	D	A	C	B	D	A	B	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	D	A	C	D	B	A	D	C	B	A

Q	21	22	23	24	25	26	27	28	29	30
Ans	D	B	A	C	D	B	D	B	C	D

Q	31	32	33	34	35	36	37	38	39	40
Ans	B	A	C	D	B	A	D	B	C	A

Q	41	42	43	44	45	46	47	48	49	50
Ans	D	B	C	A	D	B	C	A	D	B

Explanations

Q1 — C. `(A1:A5>10)` → {0,1,0,1,0}; multiplied by B → {0,2,0,4,0} → sum **6**. Note `10 > 10` is FALSE — strictly greater.

Q2 — B. 0.75 formatted as `"0.0%"` → **75.0%** (the format string controls the display, not the stored value).

Q3 — D. TRIM removes leading/trailing spaces **and collapses internal runs to a single space** → "Hello World" → **11**.

Q4 — A. 50 is **not** > 50, so the FALSE branch runs: 200 × 0.05 = **10**.

Q5 — C. Section breaks let one document have different headers, page numbering (e.g. roman then arabic) and orientation/ margins per section.

Q6 — B. Slide Sorter shows thumbnails for reordering, transitions and timings. Normal/Outline view is for editing content.

Q7 — D. Bcc recipients are hidden from all other recipients. Cc is visible to everyone.

Q8 — A. A scatter (XY) plot shows the relationship between two numeric variables; column/bar compares categories, pie shows parts of a whole.

Q9 — B. Need 2^h – 2 ≥ 100 → h = 7 → 32 – 7 = **/25** (126 usable hosts). /26 gives only 62.

Q10 — C. 4 subnets = 2 extra bits → /26 (block 64) → 10.0.0.0/26, **10.0.0.64/26**, .128/26, .192/26.

Q11 — D. /28 = block 16 → 192.168.1.0–.15; `.0` is the network address, `.15` the broadcast → usable `.1`–`.14`.

Q12 — A. `::1/128` is loopback. `FE80::/10` link-local, `FF02::1` all-nodes multicast, `2001:db8::/32` documentation.

Q13 — C. Link-local = **FE80::/10** (auto-configured, never routed).

Q14 — D. NDP (ICMPv6) handles address resolution, router discovery and duplicate-address detection; ARP does not exist in IPv6.

Q15 — B. SFTP = SSH File Transfer Protocol (port 22). FTPS = FTP over TLS (990/989); TFTP is trivial/unauthenticated UDP.

Q16 — A. UDP is **connectionless** — that is the false statement. All the others are true.

Q17 — D. A firewall filters on L3/L4 (IP, protocol, port) — and an NGFW/application firewall also inspects L7. A L2 switch only sees MAC addresses.

Q18 — C. Auto-scaling (horizontal elasticity) adds/removes instances by policy. Load balancing only distributes existing traffic.

Q19 — B. MFA defeats replay of a stolen password. Complexity rules and timeouts do not stop a valid leaked credential being replayed.

Q20 — A. Type 1 = bare metal (ESXi, Hyper-V, KVM) directly on hardware. Type 2 = hosted (VirtualBox, VMware Workstation) on top of an OS.

Q21 — D. A dynamic array's `append`/`push` is O(1) amortised; the occasional resize costs O(n).

Q22 — B. Insertion sort is O(n) on nearly sorted data (adaptive) — it only shifts each element a short distance. Quick sort with a bad pivot degrades to $O(n^2)$.

Q23 — A. Push opens, pop closes, and a mismatch/leftover stack means unbalanced.

Q24 — C. 31 = 2■ – 1 → a perfect tree with levels 0–4; leaves live on level 4 → 2■ = **16**. (General rule: leaves = ■n/2■ for a complete tree.)

Q25 — D. BCNF is stricter: it forbids *every* non-trivial dependency whose determinant is not a superkey, so BCNF ⊂ 3NF.

Q26 — B. A■ = {A,B,C,D} (A→B→C→D) → **{A}** is the (minimal) candidate key.

Q27 — D. A: 2 orders / 400; B: 2 orders / 450; C: 1 / 50 (fails HAVING). Ties on `c` → `s DESC` → **B (450)** first, then A.

Q28 — B. Isolation = concurrent transactions appear to run serially (serialisable). Atomicity = all-or-nothing; Durability = survives crashes; Consistency = valid states.

Q29 — C. The base constructor runs first and calls `show()`, which dispatches **virtually** to `B.show()` → prints **B**. (This is why calling overridable methods from a constructor is dangerous: B's fields are not initialised yet.)

Q30 — D. `<nav>` is the semantic element for primary navigation. `<menu>` is for toolbars/context menus.

Q31 — B. `map` → [2,4,6]; `filter(x > 2)` → **[4, 6]**. Neither method mutates the original array.

Q32 — A. 2■ = 256 < 500 ≤ 512 = 2■ → **9 bits**.

Q33 — C. Evens 2 + 4 = 6; odds 1 + 3 + 5 = 9 → `6 9`.

Q34 — D. 7 + 6 + 5 + 4 = **22** (recursion peels the last digit off each time).

Q35 — B. `f(3)` → x = 13, returns 13. `f(5)` → x = 18, returns 18. 13 + 18 = **31**. The global persists between calls.

Q36 — A. Consecutive equal pairs: "ss" (2–3), "ss" (5–6), "pp" (8–9) → **3**.

Q37 — D. (1,10)→(3,9)→(5,8)→(7,7); at (7,7) the condition fails → `c = 3, i = 7, j = 7`.

Q38 — B. Reversed array = [6,5,4,3,2,1]; `A[1] = 5`, `A[4] = 2` → `5 2`.

Q39 — C. F(0)=0, F(1)=1, F(2)=2, F(3)=5, F(4)=2×5+2 = **12**.

Q40 — A. Minima: 11 (swap 1), 12 (swap 2), 22 (swap 3), then 25 is already in place → **3** swaps. Selection sort is O(n²) comparisons but only O(n) swaps.

Q41 — D. 100 → 33 → 11 → 3 → 1 = **4** divisions, ending with `n = 1`.

Q42 — B. "acCenture": a(0) c(1) **C(2)** → returns **2** (the first uppercase letter, 0-based).

Q43 — C. C[0][0] = 1×5 + 2×7 = 19; C[1][1] = 3×6 + 4×8 = 50 → `19 50`.

Q44 — A. Sum 1…10 = 55, minus 4 and 8 → **43**. The `CONTINUE` skips only the addition.

Q45 — D. i = 0: a vs a ✓; i = 1: b vs b ✓; i = 2: **x vs c ✗** → FALSE and BREAK. The loop runs to `n DIV 2 – 1` = 2.

Q46 — B. moves(n) = 2■ – 1 → 2■ – 1 = **31**.

Q47 — C. Inner query gives 70000; the outer max of salaries **strictly less** than 70000 = **60000** (the classic "second highest" pattern).

Q48 — A. TRUNCATE is DDL, deallocates pages, is minimally logged and takes no WHERE clause. DELETE is DML, logged row-by-row and supports WHERE; DROP removes the table definition itself.

Q49 — D. Not a reverse! i = 0 → arr[0] = arr[2] = 3; i = 1 → arr[1] = arr[1] = 2; i = 2 → arr[2] = arr[0] = **3** → `323`.

Q50 — B. `innerHTML` parses the string as HTML → **Hi** in bold. `textContent` would display the literal `<b>Hi</b>`. The script sits *after* the element, so the element exists.

Score guide (Set 3)

Score	Verdict
43–50	Excellent — safely above Accenture level
34–42	Good — revise the missed areas
25–33	Needs work — revisit the notes, retry after 3 days
< 25	Return to chapters 01–06 before Set 4